

Shuquan XIAO

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EDUCATION

Master of Physics, Université Paris-Saclay, ENS Paris-Saclay, France

Sep.2024 – Jul.2025

➤ M2: Physics, Engineering and Chemistry for Biology

Bachelor of Engineering, Harbin Institute of Technology-Shenzhen, China

Sep.2020 - Jul.2024

- Major in Opto-electronic Information of Science and Engineering, GPA: 87.272/100, Rank: 4/21 (top 20%)
- Core Courses: Introduction to Electronic Information Engineering, Engineering Optics, Signals and Systems, Introduction to Machine Learning, Principles of Lasers, Photoelectric Detection Technology, Electromagnetics, Optoelectronic Device & Technology, Comprehensive Design of Optoelectronic Information Technologies, Waveguide Optics, Optical Precise Measurement Technology, Optical Design Experiment, etc.

PUBLICATIONS & CONFERENCE PRESENTATION

1. **Shuquan Xiao**, Yunxu Sun, Martha Vardaki, Wei Liu*, “Theoretical framework for calibrating the depth-dependent optical scattering in layered human skin using spatially-offset measurements”, **Optics Letters**, 2024, DOI: doi.org/10.1364/OL.532793 (selected for **Optica Open Highlight**)
2. **Shuquan Xiao**, Yunxu Sun, Ting Ma, Zhengjun Liu, and Wei Liu*, “A Monte-Carlo simulation study of the depth-resolved backscattering photons in layered Human skin”, Oral, **16th International Conference on Photonics and Imaging in Biology and Medicine**, 2023, Haikou, HN, China

RESEARCH ACTIVITIES

➤ **03/2025 – 07/2025, Master Thesis, European Molecular Biology Laboratory**

Project: Photoacoustic and Wavefront Shaping

Supervisor: Robert Prevedel, group leader

Co-supervisor: Sylvain Gigan, professor (Laboratoire Kastler Brossel)

Contents: Aiming to develop a non-invasive method for diffraction-limited fluorescence imaging, at deep depth in scattering tissues, the project proposed a hybrid method - applying photoacoustic feedback in wavefront shaping and computational fluorescence imaging. The thesis finished the simulation work of the whole imaging process, and successfully achieved several hundred micrometer imaging through biological tissues.

➤ **09/2023 – 05/2024, Undergraduate Thesis, HITSZ Optical Imaging Laboratory**

Project: Research on Denoised Methods of Optical Coherent Diffractive Imaging

Supervisor: Wei Liu, associate professor & Lab PI

Contents: Coherent Diffractive Imaging(CDI) is a lens-less imaging technique. In this project, based on the investigations of the CDI principle, the denoise methods, and the reconstruction algorithms, the analysis of the noise source and the optimization of the reconstruction algorithm were proposed to solve the noise problem in CDI. The thesis finished the simulation of CDI and the optimization of Phase Retrieval Algorithm

by simulation and experiments, and utilizing deep learning, proposed a denoised net for CDI, which improved the imaging resolution and accuracy.

➤ **08/2022 – 08/2023, Project Leader, HITSZ Optical Imaging Laboratory**

Project: Theoretical framework for calibrating the depth-dependent optical scattering in layered human skin using spatially-offset measurements

Supervisor: Wei Liu, associate professor & Lab PI

Contents: In this project, we reported a Monte-Carlo-based quantification modeling platform combined with a novel scattering spectrum decomposition method to explore the depth-dependent optical scattering contributions in Human skin. The simulation result indicated that the layer-dependent scattering contribution was compositely encoded into the spatially-offset measurements and varied with the illumination incidence angle. For calibrating the layer-dependent scattering contribution, a modified nonlinear independent component processing algorithm was applied to the spatially-offset measurements by decomposing the photon weights of each layer. The calibration results figured out the major scattering contribution of each layer along the offset axis under different incidence angles.

INTERNSHIPS

➤ **03/2025 – 07/2025**, Europe Molecular Biology Laboratory, Heidelberg
Master Thesis, Photoacoustic and Wavefront Shaping.

➤ **03/2024 – 05/2024**, Harbin Institute of Technology, Shenzhen
Teaching assistant of Physical Optics - Experiment

➤ **09/2023 – 01/2024**, Harbin Institute of Technology, Shenzhen
Teaching assistant of Optical Design Experiment

AWARDS

1. 06/2024, Outstanding Undergraduate Thesis
2. 05/2024, Outstanding Undergraduate Graduate
3. 11/2023, Outstanding Student Leader Award
4. 11/2023, The Second-class Academic Scholarship in the 2022-2023 Academic Year (top 15%)
5. 12/2022, National First Prize in the 8th National Collegiate Physics Experiment Competition
6. 11/2022, The Second-class Academic Scholarship in the 2021-2022 Academic Year (top 15%)
7. 11/2022, Outstanding Student Leader Award
8. 07/2022, Second prize in the 1st Harbin Institute of Technology (Shenzhen) Optoelectronic Design Competition
9. 05/2022, Meritorious Winner in The Mathematical Contest in Modeling (MCM)
10. 12/2021, Outstanding Student Award

SELF-ASSESSMENT

➤ **Skills**

Programming: Matlab, Python, C

Software Skills: SOLIDWORKS, VirtualLab Fusion simulation, Candence, Vivado&Verilog, Code Composer Studio